

# Lorenzo Corti

Residence: Netherlands – Nationality: Italian

🏠 <https://lcorti.github.io> — ✉ [corti.lorenz@gmail.com](mailto:corti.lorenz@gmail.com) — [🌐 linkedin.com/in/lorenzo-corti-46a22b181](https://www.linkedin.com/in/lorenzo-corti-46a22b181) — [🐙 github.com/LCorti](https://github.com/LCorti)

## Summary

I work on **tools** and **methods** for diagnosing large **vision and language models**. My work sits at the intersection of **NLP** and **explainable AI**, where I develop rigorous **evaluations** and integrate modern vision and language models into end-to-end **applications**. I have hands-on experience with frameworks such as **PyTorch**, **Hugging Face Transformers**, and **spaCy**, and I am particularly interested in building reliable and **robust multimodal systems** and workflows.

I am currently pursuing a PhD at Delft University of Technology. Previously, I received my MSc and BSc in Computer Science and Engineering from Politecnico di Milano.

## Skills

**Languages** Python, Java, C, Javascript  
**Data Management** Relational DBs and NoSQL  
**Web Development** Spring, FastAPI, Django  
**DevOps** Docker, VPS, Git

**Machine Learning** NumPy, Pandas, scikit-learn, spaCy  
**Deep Learning** Pytorch, Captum, Huggingface  
Transformers, Datasets, and Evaluate  
**Project Management** Research design and planning

## Education

**Delft University of Technology, Delft, NL**

**Feb 2022 – 2027\***

*PhD Candidate in Computer Science*

*\*Finalising my dissertation and can terminate earlier than Feb. 2027.*

- *Dissertation:* Beyond Benchmarks – Diagnosis of Vision and Language Model Behaviours
- *Supervision:* Prof.Dr.Ir. Geert-Jan Houben (promotor) and Dr. Jie Yang (copromotor)

**Politecnico di Milano, Milan, IT**

*Master of Science in Computer Science and Engineering (110/110)*

*Sept 2017 – Dec 2019*

- *Thesis:* Time Conditional Generative Adversarial Networks for Augmentation of Irregularly Sampled Time Series
- *Supervision:* Marco Brambilla (Politecnico di Milano) and Pavlos Protopapas (Harvard University)

*Bachelor of Science in Computer Science and Engineering (98/110)*

*Sept 2014 – July 2017*

- *Thesis Project:* Development of a customer relationship management system for an advertising company.
- *Deliverables:* Requirement analysis with i\* framework, UML design, and documented codebase.

## Professional Experience

**Doctoral Candidate**

**Feb 2022 – 2027\***

*Delft University of Technology, Delft, NL*

- **Diagnosis of Vision Language Model Behaviours:** Design, implementation, and evaluation of a method to diagnose the behaviours of VLMs. Observable behaviours are verified against specifications of desired behaviours. Concept-level contributions towards VLM outputs are quantified using a Double-ML causal estimator fitted to synthetically occluded visual inputs. Applied to 4 VLMs across 4 benchmarks for multiple-choice and open-ended visual question answering.  
🔗 *Implementation:* Hugging Face Transformers, PyTorch, and NumPy to self-host and work with VLMs, Hugging Face Datasets, Pandas, OpenCV, SentenceTransformers, spaCy, and scikit-learn for data preprocessing and analysis, IETrans and NetworkX for scene graph extraction and analysis, DoWhy and EconML for causal analysis, and Matplotlib and Seaborn for visualisation.
- **Toolkit for LLM Behaviour Diagnosis:** Design and development of a toolkit to diagnose LLM behaviours on known linguistic structures. The toolkit supports encoder-only, encoder-decoder, and decoder-only transformer-based LLMs, different NLP datasets, known NLP metrics, inspection and tagging of LLM outputs, and behavioural resting and reporting. The tool is externally validated with NLP industry practitioners.  
🔗 *Implementation:* Hugging Face Transformers and Datasets, PyTorch, spaCy, and NumPy as foundation, Pandas and scikit-learn for metrics computation, Captum for interpretable ML support, FastAPI and MongoDB for processing backend, Django and MySQL for frontend, and Docker for containerisation and self-hosting.
- **Extraction of Behaviours of Vision Models:** Explainable AI method to extract observable behaviours of CNN-based vision models. Applied the method to vision models in the presence of known biases (introduced via fine-tuning) and evaluated in terms of fidelity, coherence, and accuracy of the behaviours extracted.  
🔗 *Implementation:* DoWhy for causal estimates, CDT and NetworkX for causal graphs discovery and analysis, NumPy, Pandas, scikit-learn, Matplotlib for data analysis and visualisation.
- Additionally, under the TU Delft AI Lab Design@Scale, I have mentored 8 M.Sc. theses, 5 B.Sc. theses, and 10 student group projects on NLP, language models, model interpretability, human computation, and information systems.

- **GANs for Time Series Imputation:** Continued training from scratch of GANs combining LSTM and Convolutional networks. Implemented different training regimes and normalisations to prevent mode collapse.  
 </> Implementation: TensorFlow, NumPy, scikit-learn.
- **IDiOM:** Modular and open-source NLP data analysis pipeline for disinformation. Implements data collection from social and news media sources and runs natural language processing and machine learning analyses.  
 </> Implementation: NumPy, scikit-learn, Hugging Face Transformers, spaCy, MongoDB, and custom data ingestion modules.
- **VaccinItaly:** Analysis of the Italian discourse on COVID-19 (perception and vaccine intake/hesitancy) on social media (Twitter and Facebook) and news outlets. Includes temporal, geographical and lexical analyses as well as training of stance classifiers (pro- or anti-vaccination).  
 </> Implementation: IDiOM pipeline (above) and self-developed code with NumPy, scikit-learn, spaCy, and Matplotlib.
- **COCTEAU:** Gamified, crowdsourcing platform aiming to enable large-scale citizen engagement.  
 </> Implementation: Spring Boot, Java, jQuery, Bootstrap, and MySQL.
- **PERSEUS:** Document search and analysis platform leveraging language modelling for NER, topic extraction, and document analysis.  
 </> Implementation: Python FastAPI, Hugging Face Transformers, and scikit-learn backend, jQuery and Bootstrap frontend, and Elasticsearch storage.

## Languages

---

- **Italian**, Native.
- **English**, Full Professional Proficiency.
- **Spanish**, Elementary.

## Selected Publications

---

1. *DiaVLo: Diagnosing Behaviours of Vision-Language Models*. **Lorenzo Corti**, Jie Yang. To appear in Findings of the Association for Computational Linguistics: EMNLP 2026.
2. *Toolkit for LLM Behaviour Diagnosis*. **Lorenzo Corti**, Sebastiaan Beekman, Jie Yang. *Under review*.
3. *A.I. Robustness: a Human-Centered Perspective on Technological Challenges and Opportunities*. Andrea Tocchetti\*, **Lorenzo Corti\***, Agathe Balayn\*, Mireia Yurrita, Philip Lippmann, Marco Brambilla, Jie Yang (\* equal contribution). In: ACM Computing Surveys. 2025. DOI: <https://doi.org/10.1145/3665926>.
4. *CHIME: Causal Human-in-the-Loop Model Explanations*. Shreyan Biswas, **Lorenzo Corti**, Stefan Buijsman, Jie Yang. In: Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. 2022. DOI: <https://doi.org/10.1609/hcomp.v10i1.21985>.
5. *A Web-Based Co-Creation and User Engagement Method and Platform*. Andrea Tocchetti, **Lorenzo Corti**, Marco Brambilla, Diletta Di Marco. In: Proceedings of International Conference on Web Engineering. 2021. DOI: [https://doi.org/10.1007/978-3-030-74296-6\\_38](https://doi.org/10.1007/978-3-030-74296-6_38).

## Workshop and Demo Papers

1. *ARTIST: ARTificial Intelligence for Simplified Text*. **Lorenzo Corti**, Jie Yang. In: Generative AI and HCI workshop at CHI. 2023. DOI: <https://doi.org/10.48550/arXiv.2308.13458>.
2. *A content-based approach for the analysis and classification of vaccine-related stances on Twitter: the Italian scenario*. Marco Di Giovanni, **Lorenzo Corti**, Silvio Pavanetto, Francesco Pierri, Andrea Tocchetti, Marco Brambilla. In: Workshop Proceedings of the 15th International AAAI Conference on Web and Social Media. 2021. DOI: <https://doi.org/10.36190/2021.52>.